

**EEE2047S INFORMATION SHEET**  
**(version 2025-10-26)**

**Convolution**

$$x(t) * y(t) = \int_{-\infty}^{\infty} x(\tau)y(t - \tau) d\tau$$

**Fourier Series**

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \quad \text{where} \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

**Parseval's theorem for periodic signals**

$$P_{ave} = \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |c_k|^2$$

**Fourier Transform**

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad \text{where} \quad X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

**If  $x(t)$  is periodic:**

$$X(\omega) = \sum_{k=-\infty}^{\infty} 2\pi c_k \delta(\omega - k\omega_0) \quad (\text{for periodic signals})$$

$$\text{If } x(t) = z(t) * \delta_T(t) \text{ where } \delta_T(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT) \text{ then } c_k = \frac{1}{T} Z(k\omega_0)$$

**Transfer Function of a LCCDE**

$$a_0 y(t) + a_1 \frac{dy}{dt} + a_2 \frac{d^2 y}{dt^2} + \dots = b_0 x(t) + b_1 \frac{dx}{dt} + b_2 \frac{d^2 x}{dt^2} + \dots$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{b_0 + b_1 j\omega + b_2 (j\omega)^2 + \dots}{a_0 + a_1 j\omega + a_2 (j\omega)^2 + \dots} = \frac{\sum_k b_k (j\omega)^k}{\sum_m a_m (j\omega)^m}$$

## Fourier transform properties

| Property                             | Transform Pair/Property                                                                                         |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Linearity                            | $ax(t) + bv(t) \longleftrightarrow aX(\omega) + bV(\omega)$                                                     |
| Complex conjugate                    | $x^*(t) \longleftrightarrow X^*(-\omega)$                                                                       |
| Time shift                           | $x(t - c) \longleftrightarrow X(\omega)e^{-j\omega c} \quad (c \text{ any real number})$                        |
| Time scaling                         | $x(at) \longleftrightarrow \frac{1}{ a }X\left(\frac{\omega}{a}\right) \quad (a \text{ any real number})$       |
| Time reversal                        | $x(-t) \longleftrightarrow X(-\omega)$                                                                          |
| Multiplication by power of $t$       | $t^n x(t) \longleftrightarrow j^n \frac{d^n}{d\omega^n} X(\omega) \quad n = 1, 2, \dots$                        |
| Frequency shift                      | $x(t)e^{j\omega_0 t} \longleftrightarrow X(\omega - \omega_0) \quad \omega_0 \text{ real}$                      |
| Multiplication by $\cos(\omega_0 t)$ | $x(t) \cos(\omega_0 t) \longleftrightarrow \frac{1}{2}[X(\omega + \omega_0) + X(\omega - \omega_0)]$            |
| Differentiation in time domain       | $\frac{d^n}{dt^n} x(t) \longleftrightarrow (j\omega)^n X(\omega) \quad n = 1, 2, \dots$                         |
| Integration                          | $\int_{-\infty}^t x(\lambda) d\lambda \longleftrightarrow \frac{1}{j\omega} X(\omega) + \pi X(0)\delta(\omega)$ |
| Convolution in time domain           | $x(t) * v(t) \longleftrightarrow X(\omega)V(\omega)$                                                            |
| Multiplication in time domain        | $x(t)v(t) \longleftrightarrow \frac{1}{2\pi} X(\omega) * V(\omega)$                                             |
| Parseval's theorem                   | $\int_{-\infty}^{\infty} x^*(t)v(t)dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega)V(\omega)d\omega$     |
| Parseval's theorem (special case)    | $\int_{-\infty}^{\infty}  x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty}  X(\omega) ^2 d\omega$            |
| Duality                              | $X(t) \longleftrightarrow 2\pi x(-\omega)$                                                                      |
| Conjugate symmetry for real signals: | If $x(t)$ is real-valued, then $X(-\omega) = X^*(\omega)$                                                       |

## Common Fourier Transform Pairs

|                                                                                                                          |                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t} d\omega$                                           | $X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$                                                                      |
| 1 $(-\infty < t < \infty)$                                                                                               | $2\pi\delta(\omega)$                                                                                                             |
| $-0.5 + u(t)$                                                                                                            | $\frac{1}{j\omega}$                                                                                                              |
| $u(t)$                                                                                                                   | $\pi\delta(\omega) + \frac{1}{j\omega}$                                                                                          |
| $\delta(t)$                                                                                                              | 1                                                                                                                                |
| $\delta(t - c)$                                                                                                          | $e^{-j\omega c} \quad (c \text{ any real number})$                                                                               |
| $e^{-bt}u(t)$                                                                                                            | $\frac{1}{j\omega + b} \quad (b > 0)$                                                                                            |
| $e^{j\omega_0 t}$                                                                                                        | $2\pi\delta(\omega - \omega_0) \quad (\omega_0 \text{ any real number})$                                                         |
| $p_\tau(t)$ where $p_\tau(t) = \text{rect}(t/\tau)$                                                                      | $\tau \text{sinc}\left(\frac{\tau\omega}{2\pi}\right) = \tau \text{Sa}\left(\frac{\tau\omega}{2}\right)$                         |
| $\tau \text{sinc}\left(\frac{\tau t}{2\pi}\right) = \tau \text{Sa}\left(\frac{\tau t}{2}\right)$                         | $2\pi p_\tau(\omega) \quad \text{where } p_\tau(\omega) = \text{rect}(\omega/\tau)$                                              |
| $\left(1 - \frac{2 t }{\tau}\right)p_\tau(t)$                                                                            | $\frac{\tau}{2} \text{sinc}^2\left(\frac{\tau\omega}{4\pi}\right) = \frac{\tau}{2} \text{Sa}^2\left(\frac{\tau\omega}{4}\right)$ |
| $\frac{\tau}{2} \text{sinc}^2\left(\frac{\tau t}{4\pi}\right) = \frac{\tau}{2} \text{Sa}^2\left(\frac{\tau t}{4}\right)$ | $2\pi \left(1 - \frac{2 \omega }{\tau}\right)p_\tau(\omega)$                                                                     |
| $\cos(\omega_0 t + \theta)$                                                                                              | $\pi[e^{-j\theta}\delta(\omega + \omega_0) + e^{j\theta}\delta(\omega - \omega_0)]$                                              |
| $\sin(\omega_0 t + \theta)$                                                                                              | $j\pi[e^{-j\theta}\delta(\omega + \omega_0) - e^{j\theta}\delta(\omega - \omega_0)]$                                             |
| $\sum_{n=-\infty}^{\infty} \delta(t - nT)$                                                                               | $\frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\frac{2\pi}{T})$                                                      |

where

$$p_\tau(t) = u(t + \tau/2) - u(t - \tau/2)$$

$$\text{rect}(t) = u(t + 1/2) - u(t - 1/2)$$

$$p_\tau(t) = \text{rect}(t/\tau)$$

$$\text{Sa}(\lambda) = \sin(\lambda)/\lambda$$

$$\text{sinc}(\lambda) = \sin(\pi\lambda)/(\pi\lambda)$$

$$\text{Sa}(\lambda) = \text{sinc}(\lambda/\pi) \quad \text{sinc}(\lambda) = \text{Sa}(\pi\lambda)$$

## Laplace transform properties

| Property                  | Transform Pair/Property                                                                                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Linearity                 | $ax(t) + bv(t) \longleftrightarrow aX(s) + bV(s)$                                                                                                                                                        |
| Time shift                | $x(t-a)u(t-a) \longleftrightarrow e^{-as}X(s) \quad a \geq 0$                                                                                                                                            |
| Time scaling              | $x(at) \longleftrightarrow \frac{1}{a}X\left(\frac{s}{a}\right) \quad a > 0$                                                                                                                             |
| Frequency differentiation | $t^n x(t) \longleftrightarrow (-1)^n X^{(n)}(s)$                                                                                                                                                         |
| Frequency shift           | $e^{at}x(t) \longleftrightarrow X(s-a)$                                                                                                                                                                  |
| Differentiation           | $x'(t) \longleftrightarrow sX(s) - x(0^-)$<br>$x''(t) \longleftrightarrow s^2X(s) - sx(0^-) - x'(0^-)$<br>$x^{(n)}(t) \longleftrightarrow s^nX(s) - s^{n-1}x(0^-) - \dots - x^{(n-1)}(0^-)$              |
| Integration               | $\int_{0^-}^t x(\lambda)d\lambda \longleftrightarrow \frac{1}{s}X(s)$<br>$\int_{-\infty}^t x(\lambda)d\lambda \longleftrightarrow \frac{1}{s}X(s) + \frac{1}{s} \int_{-\infty}^{0^-} x(\lambda)d\lambda$ |
| Time convolution          | $x(t) * v(t) \longleftrightarrow X(s)V(s)$                                                                                                                                                               |
| Frequency convolution     | $x(t)v(t) \longleftrightarrow \frac{1}{2\pi j}X(s) * V(s)$                                                                                                                                               |

Initial value:  $f(0^+) = \lim_{s \rightarrow \infty} sF(s)$

Final value:  $f(\infty) = \lim_{s \rightarrow 0} sF(s)$  with all poles in left-hand plane

## Common Unilateral Laplace Transform Pairs

| $x(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(s)e^{st}ds$                                       | $X(s) = \int_0^\infty x(t)e^{-st}dt$                                                   |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| $\delta(t)$                                                                                               | 1                                                                                      |
| $u(t)$                                                                                                    | $\frac{1}{s}$                                                                          |
| $tu(t)$                                                                                                   | $\frac{1}{s^2}$                                                                        |
| $t^n u(t)$                                                                                                | $\frac{n!}{s^{n+1}}$                                                                   |
| $e^{\lambda t}u(t)$                                                                                       | $\frac{1}{s-\lambda}$                                                                  |
| $te^{\lambda t}u(t)$                                                                                      | $\frac{1}{(s-\lambda)^2}$                                                              |
| $t^n e^{\lambda t}u(t)$                                                                                   | $\frac{n!}{(s-\lambda)^{n+1}}$                                                         |
| $\cos(bt)u(t)$                                                                                            | $\frac{s}{s^2+b^2}$                                                                    |
| $\sin(bt)u(t)$                                                                                            | $\frac{b}{s^2+b^2}$                                                                    |
| $e^{-at} \cos(bt)u(t)$                                                                                    | $\frac{s+a}{(s+a)^2+b^2}$                                                              |
| $e^{-at} \sin(bt)u(t)$                                                                                    | $\frac{b}{(s+a)^2+b^2}$                                                                |
| $re^{-at} \cos(bt+\theta)u(t)$                                                                            | $\frac{(r \cos \theta)s + (ar \cos \theta - br \sin \theta)}{s^2 + 2as + (a^2 + b^2)}$ |
| $re^{-at} \cos(bt+\theta)u(t)$                                                                            | $\frac{0.5re^{j\theta}}{s+a-jb} + \frac{0.5re^{-j\theta}}{s+a+jb}$                     |
| $re^{-at} \cos(bt+\theta)u(t)$                                                                            | $\frac{As+B}{s^2+2as+c}$                                                               |
| $r = \sqrt{\frac{A^2c+B^2-2ABa}{c-a^2}}, b = \sqrt{c-a^2}, \theta = \tan^{-1} \frac{Aa-b}{A\sqrt{c-a^2}}$ |                                                                                        |

## Trigonometric identities

$$\sin(-\theta) = -\sin(\theta) \quad \cos(-\theta) = \cos(\theta) \quad \tan(-\theta) = -\tan(\theta)$$

$$\sin^2(\theta) + \cos^2(\theta) = 1$$

$$\sin(2\theta) = 2\sin(\theta)\cos(\theta) \quad \cos(2\theta) = \cos^2(\theta) - \sin^2(\theta) = 2\cos^2(\theta) - 1 = 1 - 2\sin^2(\theta)$$

$$\sin(\theta_1 + \theta_2) = \sin(\theta_1)\cos(\theta_2) + \cos(\theta_1)\sin(\theta_2) \quad \cos(\theta_1 + \theta_2) = \cos(\theta_1)\cos(\theta_2) - \sin(\theta_1)\sin(\theta_2)$$

$$e^{j\theta} = \cos(\theta) + j\sin(\theta)$$

$$\cos(\theta) = \frac{1}{2}e^{j\theta} + \frac{1}{2}e^{-j\theta} \quad \sin(\theta) = \frac{1}{2j}e^{j\theta} - \frac{1}{2j}e^{-j\theta}$$

## Complex Numbers

$$zz^* = |z|^2 \quad 1/z = z^*/zz^* = z^*/|z|^2 \quad |xyz| = |x||y||z|$$

$$z = Ae^{j\theta} \quad |Ae^{j\theta}| = |A| |e^{j\theta}| = |A| \quad \tan \theta = \frac{\text{Im}\{z\}}{\text{Re}\{z\}}$$